European Polymer Journal 211 (2024) 113022

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Polymers with one free gallic acid per monomer unit. Controlled synthesis and intrinsic antioxidant/antibacterial properties

Ali Khalil [a], Carmen Moya Lopez [a], Francisco Carlos Pérez-Martínez [b], Christine Gerardin-Charbonnier [c], Khalid Ferji [a], Carlos Alonso Moreno [d], Jean-Luc Six [a,*]

[a] Université de Lorraine, CNRS, LCPM, F-54000 Nancy, France
[b] Facultad de Enfermería de Albacete, Universidad de Castilla-La Mancha, 02008 Albacete, Spain
[c] Université de Lorraine, INRAE, LERMAB, F-54000 Nancy, France
[d] Departamento de Química Inorgánica, Orgánica Y Bioquímica-Centro de Innovación en Química Avanzada (ORFEO-CINQA), Unidad nanoDrug, Facultad de Farmacia, Universidad de Castilla-La Mancha, 02008 Albacete, Spain

ARTICLE INFO

Keywords:
Polyphenol
Bio-based monomer
Photo-RAFT
Controlled polymerization
Free radical scavenging
Bacteria

ABSTRACT

The development of bio-based thermoplastics is rapidly growing due to the depletion of fossil fuel reserves but also in the hope of obtaining polymers with original properties. Herein, we report the controlled synthesis of original homopolymers derived from gallic acid, one of the well-known natural polyphenols that exhibits various health-promoting effects. More precisely, acrylate, methacrylate and acrylamide monomers based on a protected gallic acid are firstly obtained by a rethought multistep scheme. Then, a controlled photo-mediated RAFT (Reversible Addition Fragmentation chain Transfer) polymerization is carried out before a final deprotection. By this way, original polymers bearing a free gallic acid as a lateral substituent on each monomer unit are described. The intrinsic antibacterial and antioxidant properties of these polymers are highlighted and compared with those of gallic acid.

1. Introduction

Polyphenols are natural compounds that are characterized by having at least one aromatic ring with one to several hydroxyl groups. Gallic acid (GA), known to be one of the most important polyphenols, is obtained by hydrolyzing tannins found in plants (tea leaves, hops, oak bark…) or is extracted from various fruits and nuts [1]. GA, as other polyphenols, has already been used in food products, packaging, biomedical, cosmetics or biofertilizers applications [2,3]. GA is reported to have significant antioxidant and antibacterial properties. Additionally, health-promoting effects have been identified in case of several diseases including viral infections, cancer, … [4] However, low bioavailability and bioactivity of GA strongly limit its extended exploitation in such biomedical fields. Various systems (hydrogels [5,6], nanoparticles [7,8] and others [9,10]) loaded by GA were consequently investigated to improve bioavailability and pharmacokinetics of GA.

To enlarge applications in the framework of active packaging or edible coating among others, GA was conjugated with natural polymeric substrates such as polysaccharides, soy protein or gelatin [11–19]. Very recently, GA was grafted on epoxidized natural rubber to develop self-healing flexible temperature sensors [20]. Surprisingly, to the best of our knowledge, no grafting of GA on synthetic thermoplastics was reported to develop polymeric materials with intrinsic antioxidant or antibacterial properties.

Recently, some of us have reported [10] for the first time poly(meth) acrylate exhibiting one GA lateral substituent per monomer unit. Such polymers were produced via radical polymerization of perbenzylated GA-based acrylate and methacrylate (BnGA-A and BnGA-MA, respectively) followed by a quantitative deprotection of OH functions of such perbenzylated polymers (poly(BnGA-A) and poly(BnGA-MA)). This synthesis route required a 4-steps scheme to produce protected monomers and a heterogeneous Pd/C catalyzed deprotection (H_2 pressure = 25 bar, RT, 48 h) of the resulting polymers. Free GA-based poly(meth) acrylate were then proved to exhibit intrinsic antioxidant activity [10], which can help such poly(meth)acrylate materials against oxidative aging for instance [21]. The lack of works dealing with radical polymerization of GA-based monomers may be due to the high stability of radical from GA that exhibit 10 different mesomeric forms [19,22]. Protection of all the alcohol functions of GA was thus required prior to the radical polymerization of such GA-based monomers [10]. The use of

* Corresponding author.
E-mail address: jean-luc.six@univ-lorraine.fr (J.-L. Six).

https://doi.org/10.1016/j.eurpolymj.2024.113022
Received 3 March 2024; Received in revised form 30 March 2024; Accepted 4 April 2024
Available online 7 April 2024
0014-3057/© 2024 Elsevier Ltd. All rights reserved.

GA as a multi-functional monomer within polycondensations is extensively reported but doesn't correspond to the objectives of the present paper.

Herein, we rethought the synthesis of poly(meth)acrylate exhibiting one GA moiety per monomer unit through an advantageous route that includes a three-step synthesis of protected GA-based monomers and a faster quantitative deprotection step. More precisely, acetate groups were selected to protect the 3 OH functions of GA to obtain triacetylated GA-based acrylate (AcGA-A) and methacrylate (AcGA-MA) monomers (Scheme 1). This synthetic scheme was also applied to produce original triacetylated GA-based acrylamide (AcGA-AA). Photo-mediated RAFT (Reversible Addition Fragmentation chain Transfer) polymerization (photo-RAFT), which is an alternative to thermal RAFT one, was used to afford a control of the molecular weight of produced protected polymers (P(AcGA-AA), P(AcGA-A) and P(AcGA-MA)) [10,23–29]. RAFT, one of the Reversible Deactivation Radical Polymerizations (RDRPs) developed to overcome the limits of the free radical polymerization, was selected in the present study because it is a metal-free polymerization. This is a huge advantage on atom transfer radical polymerization (ATRP), which requires a potentially toxic transition metal preventing the use of produced materials in the biomedical domain. To the best of our knowledge, apart from our former work [10], no other controlled radical polymerization of GA-based monomers were reported in the literature. After quantitative deprotection of each GA group, P(AcGA-AA), P(AcGA-A) and P(AcGA-MA) led to free-GA based polyacrylamide, polyacrylate and polymethacrylate (P(GA-AA), P(GA-A) and P(GA-MA)) respectively. Finally, antioxidant and antibacterial activities for such deprotected polymers were estimated and compared to free GA properties.

2. Experimental section

2.1. Materials

Gallic acid (GA, >97.5 %), acetic anhydride (Ac_2O, >99 %), sulfuric acid, oxalyl chloride (ClOC-COCl, 98 %), 2-hydroxyethyl acrylate (HEA, 96 %), 2-hydroxyethyl methacrylate (HEMA, 99 %), N-hydroxyethyl acrylamide (HEAA, 97 %), triethylamine (NEt_3,>99.5 %), hydrazine hydrate (H_2N-NH_2) 2,2-diphenyl-1-(2,4,6-trinitrophenyl)hydrazyl (DPPH, 95 %) and 2,2′-Azobis(2-methylpropionitrile) (AIBN, 98 %) were used as received from Merck. Dimethyl sulfoxide (DMSO, 99.7 %, Merck) was stored with molecular sieves. Acetone (99.8 %), dichloromethane (DCM, 99.9 %), dimethylformamide (DMF, 99.8 %), methanol (99.9 %), toluene (99,8%) were purchased from Carlo Erba and used as received. Chain transfer agents (4-(propylthiocarbonothioylthio)-4-cyanopentanoic acid (CTA1) and 2-(propylthiocarbonothioylthio)-2-methylpropionoic acid (CTA2)) were prepared as previously reported [29,30]. Ethyl (2,4,6-trimethylbenzoyl) phenylphosphinate (TPO-L) was purchased from Fluorochem and used as received.

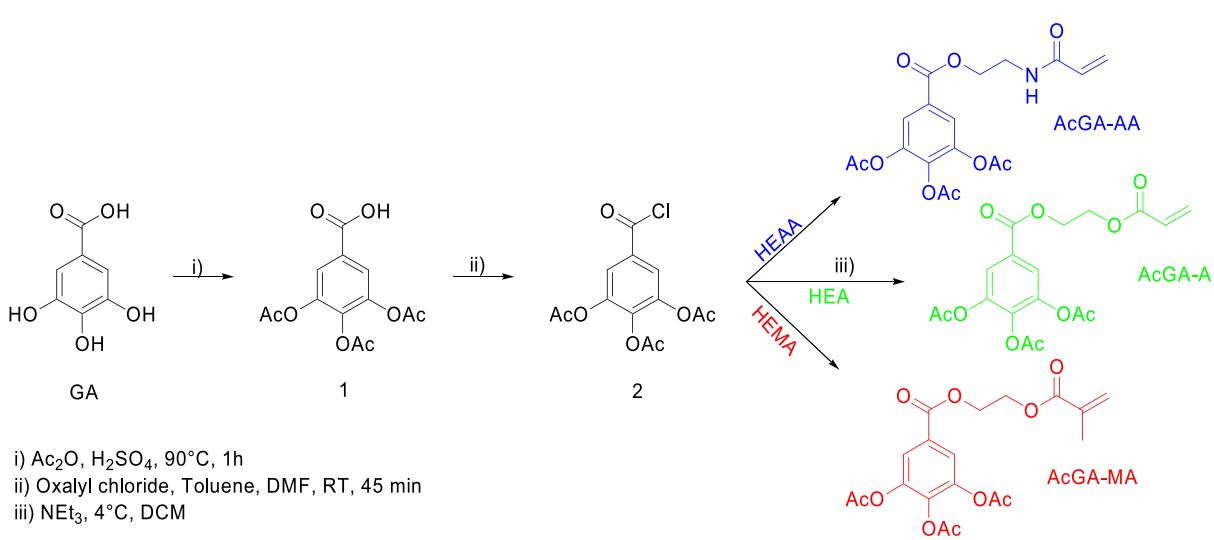
2.2. Monomer synthesis (Scheme 1)

In step i), the three hydroxyl functional groups of GA were acetylated with an excess amount of acetic anhydride in the presence of a catalytic amount of sulfuric acid, then the reaction was quenched with sodium hydrogen carbonate to obtain (1) with high yield [31]. In step ii) 20 g of (1) were dissolved in 350 mL of toluene with a catalytic amount (5 mL) of DMF. 1.1 eq. of oxalyl chloride were then slowly added at room temperature. After around 45 min, release of CO_2 bubbles was no longer observed then the reaction medium was concentrated under vacuum, but (2) was not isolated. Finally, (2) was dissolved in DCM. After cooling to 4 °C, 1.1 eq. of NEt_3 were added gradually followed by the slow addition of 1 eq. of 2-hydroxyethyl acrylamide (HEAA), 2-hydroxyethyl acrylate (HEA) or 2-hydroxyethyl (HEMA). The reaction was stirred for 30 min at 4 °C then brought to room temperature. The reaction progress was monitored by thin layer chromatography (see below for used eluents). After washing with distilled water followed by saturated $NaHCO_3$ aqueous phase, the organic phase was dried over $MgSO_4$ then concentrated under vacuum.

Pure triacetylated GA-based monomers (acrylamide (AcGA-AA), acrylate (AcGA-A) and methacrylate (AcGA-MA)) were recovered as white powders by silica gel column purification using different eluents: DCM/acetone (9/1 v/v) was used for AcGA-A (R_f = 0.62, 60 % yield) and AcGA-MA (R_f = 0.81, 55 % yield), while AcGA-AA was obtained without any further purification (yield 58 %). ^{1}H NMR spectra of such monomers are shown on Fig. 1. J-mod ^{13}C NMR and FTIR spectra are presented in supporting information (Figs. S1 and S2, respectively).

2.3. Photo-mediated RAFT polymerization

In a typical polymerization procedure, 500 mg (200 eq.) of monomer (AcGA-AA, AcGA-A or AcGA-MA) were solubilized in DMSO (1 mol/L) in a 15 mL vial, followed by the addition of 1 eq. of CTA (100 mg/mL in DMSO). The reaction mixture was purged with N_2 flow for around 10 min, then 0.1 eq. of the photoinitiator TPO-L (20 mg/mL in DMSO) was added followed by an additional purging for 3 min. The reaction mixture was finally irradiated with homemade UV–VIS LED lamp (405 nm, 10 mWcm^{-2}) [27] at room temperature to initiate the polymerization. The irradiation wavelength was fixed at 405 nm, which is suitable for TPO-L



Scheme 1. Synthesis of triacetylated GA-based acrylamide (AcGA-AA), acrylate (AcGA-A) and methacrylate (AcGA-MA).

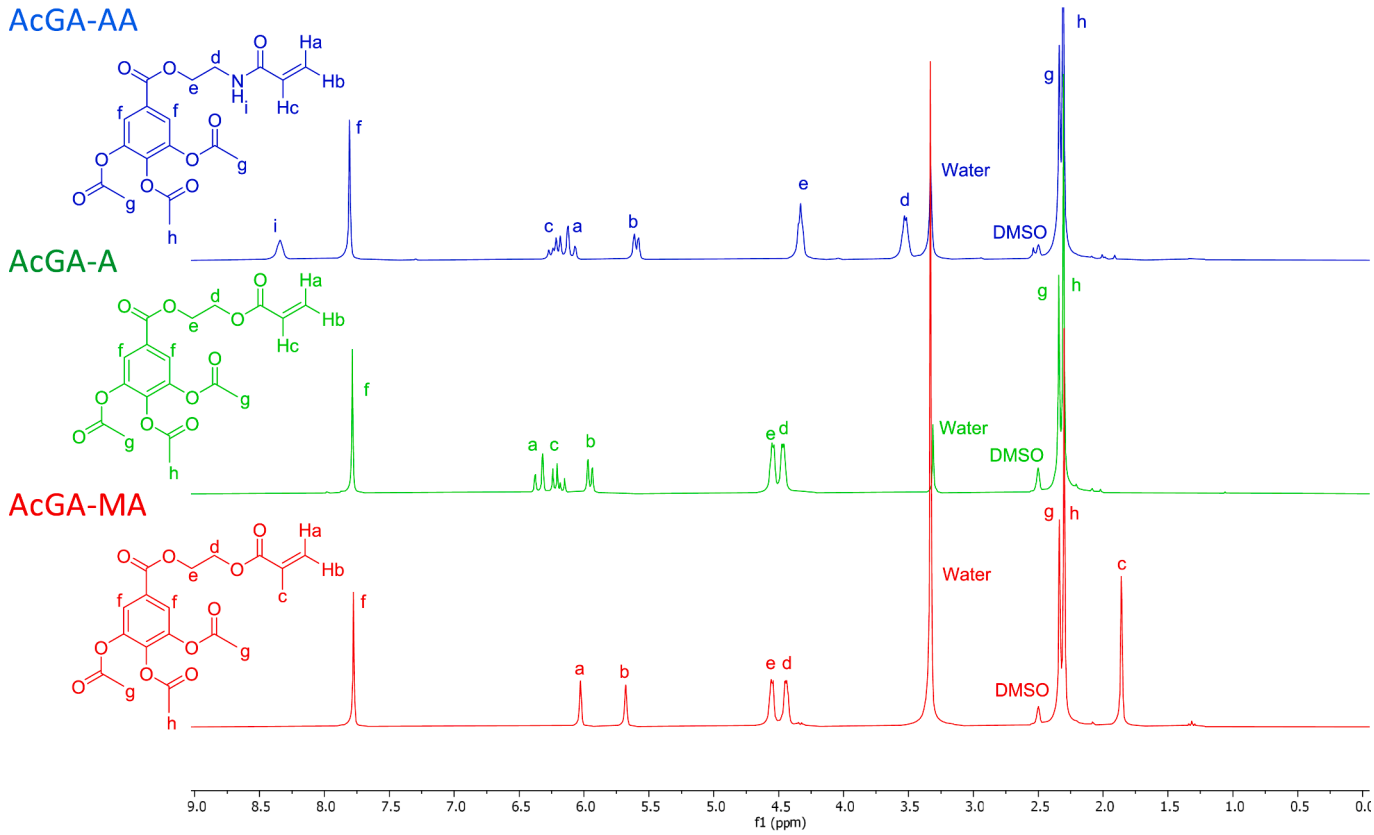


Fig. 1. ^{1}H NMR spectra (DMSO-d_6) of AcGA-AA, AcGA-A and AcGA-MA.

decomposition and the CTA stability against degradation [23,27].

Throughout polymerization, 200 μL of the reaction medium were withdrawn under nitrogen at different time and analyzed by ^{1}H NMR and SEC. Monomer conversion (x) was determined by ^{1}H NMR according to equations (1)–(3), by comparing the peaks areas of the monomer vinylic protons (from 5.5 to 6.5 ppm) with those from 7.5 to 8.5 ppm (aromatic protons and NH in case of AcGA-AA) characteristics of both monomer and polymer. Example of crude and pure P(AcGA-A) is given in Fig. S3. After polymerization, crude P(AcGA-AA), P(AcGA-A) and P (AcGA-MA) were precipitated from methanol (twice when needed), then dried under vacuum at 50ºC.

$$\text{Case of AcGA} - \text{AA}: \quad x = 1 - \left(\frac{A_{5.5-6.4\,ppm}}{3}\right) \Big/ \left(\frac{A_{7.5-8.5\,ppm}}{3}\right) \tag{1}$$

$$\text{Case of AcGA} - \text{A}: \quad \mathrm{x} = 1 - \left(\frac{A_{5.8-6.3\,ppm}}{3}\right) \Big/ \left(\frac{A_{7.5-7.8\,ppm}}{2}\right) \tag{2}$$

$$\text{Case of AcGA} - \text{MA}: \quad x = 1 - \left(\frac{A_{5.5-6.1\,ppm}}{2}\right) \Big/ \left(\frac{A_{7.5-7.8\,ppm}}{2}\right) \tag{3}$$

2.4. Deprotection step

The deprotection of P(AcGA-AA), P(AcGA-A), P(AcGA-MA) and AcGA-A was performed according to the literature [32]. Briefly, 1 g of compound was dissolved in 10 mL of DMSO at room temperature, then 3 eq. of hydrazine hydrate per GA unit were added. The reaction mixture was stirred at room temperature for 45 min, then quenched with 3 eq. of glacial acetic acid. Precipitation from acetone was performed to purify the free GA-based polymers.

In case of AcGA-A, the solution was concentrated under vacuum and the obtained solid (GA-A) was washed with distilled water, then dried under vacuum to obtain a slightly brownish solid. ^{1}H NMR spectrum is presented in Fig. S4. Upon deprotection, the signal of the acetyl groups disappeared with a significant shift of the aromatic protons from 7.8 to below 7 ppm. The broad signals of the 3 hydroxyl groups showed up from 4.7 to 5.5 ppm.

2.5. DPPH scavenging activity assay and Oxygen uptake method

DPPH scavenging activity and the antioxidation capacity of free GA-based polymers were performed according to our previous paper [10].

2.6. Antibacterial activity assay

A collection strains from the American Type Culture Collection (strains of *Staphylococcus aureus (S. aureus), Streptococcus* sp., *Escherichia coli (E. coli)*, and *Pseudomonas aeruginosa (P. aeruginosa))* were used in this study. Additionally, two clinical isolates from otitis externa patients (from the clinical Microbiology department of the Albacete University Hospital) were also used: strains of *S. aureus* and *Klebsiella aerogenes* (*K. aerogenes*) which are sensitive to various antibiotics. All strains were conserved at −80 °C until they were used with at least 24 h of culture in Tryptic Soy Agar supplemented with 5 % sheep blood (TSS, Biomérieux, France) before each experiment.

Antibacterial effects were evaluated by determining *MICs (minimum inhibitory concentrations)* for the six different strains of planktonic bacteria, according to standard methods [33]. Briefly, a broth microdilution method was determined from a 0.5 McFarland bacterial suspension previously prepared in saline serum for each strain. Thus, a series of double dilutions of compounds at concentrations of 1,000 to 0.96 mg/L

were added to the Müeller-Hinton cation-adjusted broth (CAMHB) (Sigma Aldrich, St. Louis, MO, USA) up to a final volume of 100 μL per well on a 96-well round-bottom polypropylene plate (Corning Inc., Corning, USA). Moreover, 100 μL of the bacterial suspension in CAMHB was added, in a 1:100 dilution, containing approximately 1.6×10^6 CFU/mL. A static 24-h incubation at 37 °C in 5 % CO_2 was followed. The MIC was evaluated by measuring the absorbance at 600 nm in a microtiter plate reader (Epoch TM, BioTek Instruments, Winooski, VT, USA). As negative absorbance controls, it was accompanied by serial dilutions of the compounds without microorganisms. These experiments were performed at least three times (n = 3).

2.7. Instruments and measurements

^{1}H, J-mod ^{13}C NMR spectra were recorded on a Bruker Advance 300 spectrometer (300.13 MHz, 298°K) in DMSO-d_6.

Fourier-transform infrared spectroscopy (FTIR) analyses were performed on Frontier FT-IR Spectrometer Perkin Elmer with Diamond ATR module.

Molar mass distributions of polymers were estimated by size exclusion chromatography (SEC) in DMSO containing $NaNO_3$ (8.5 g/L) as eluent, at 70 °C (elution rate 0.7 mL/min), equipped with a differential refractometer detector (RID 10A, Shimadzu), a HPLC pump (LC 20AD, Shimadzu), a degasser AF (DGU – 20A3R, Shimadzu), and three PLgel 5 μL columns. dn/dc values of 0.044, 0.065 and 0.76 were estimated for P(AcGA-AA), P(AcGA-A) and P(AcGA-MA), respectively. In case of deprotected polymers, dn/dc values of 0.067, 0.057 and 0.061 were estimated for P(GA-AA), P(GA-A) and P(GA-MA), respectively.

Differential scanning calorimetry analyses (DSC) were carried out on DSC Q2000 from TA Instruments. Three cycles were successively recorded under nitrogen from − 40 °C up to 180 °C. First cycle was used to remove all thermal history from the sample. The second heating cycle was used for determining the glass transition temperature at the inflexion point. Analysis of the data was performed using Universal® Analysis 2000 (version 4.5A) software. The heating rate was 10 °C/min while the cooling rate was 5 °C/min.

3. Results and discussion

3.1. Original triacetylated GA-based monomers

Protected GA-based acrylamide and (meth)acrylates (AcGA-AA, AcGA-A and AcGA-MA) were prepared from GA and HEAA, HEA or HEMA, respectively (Scheme 1). The first step is the acetylation of the highly reactive hydroxyl groups of GA to obtain triacetylated gallic acid (AcGA, 1) according reported procedure [31]. The corresponding triacetylated-galloyl chloride (2) was then prepared by reaction of (1) and oxalyl chloride in toluene, with the presence of a catalytic amount of DMF. (2) was directly reacted with HEAA, HEA or HEMA in the presence of triethylamine. The monomers AcGA-AA, AcGA-A and AcGA-MA were purified under appropriate conditions. Their respective ^{1}H NMR spectra (Fig. 1) showed the presence of vinylic protons between 5.4 and 6.4 ppm and the peak characteristic of methyl protons of AcGA-MA is observed as a singlet at 1.8 ppm. Acetate protons appeared around 2.3 ppm when aromatic ones at 7.76 ppm. In case of (meth)acrylate monomers, the 2 sets of ester protons were observed between 4.25 and 4.75 ppm. For AcGA-AA, the NH proton appeared around 8.3 ppm, O-methylene protons at 4.3 ppm when N-methylene protons appeared at 3.5 ppm. Integration of all of these various peaks confirmed the synthesis of pure AcGA-AA, AcGA-A and AcGA-MA.

3.2. Photo-polymerization RAFT of GA-based monomers

First of all, the necessity to fully protect the OH functions of GA-based monomer was proven by carrying out the photo-mediated RAFT polymerization of GA-AA in DMSO at room temperature, using CTA2 as RAFT agent and TPO-L as photo-initiator (Scheme 2). ^{1}H NMR spectrum of aliquots taken after 2 h of polymerization showed the unsuccessful polymerization of GA-AA, where the ratio of vinylic to aromatic protons remained unchanged. This proved the ability of unprotected GA part to scavenge radicals, and thus to prevent the radical polymerization of the corresponding unprotected monomer.

In a second step, the photo-RAFT polymerizations of AcGA-AA, AcGA-A and AcGA-MA were then carried out using $[CTA]_0/[TPO\text{-}L]_0 = 10$ and $[M]_0/[CTA]_0 = 200$ initial molar ratios. The photo-RAFT polymerization kinetics was studied thanks to several aliquots taken from the reaction medium. The first-order kinetic plots *vs* time of the polymerizations performed with CTA1 or CTA2 are shown in Fig. 2.

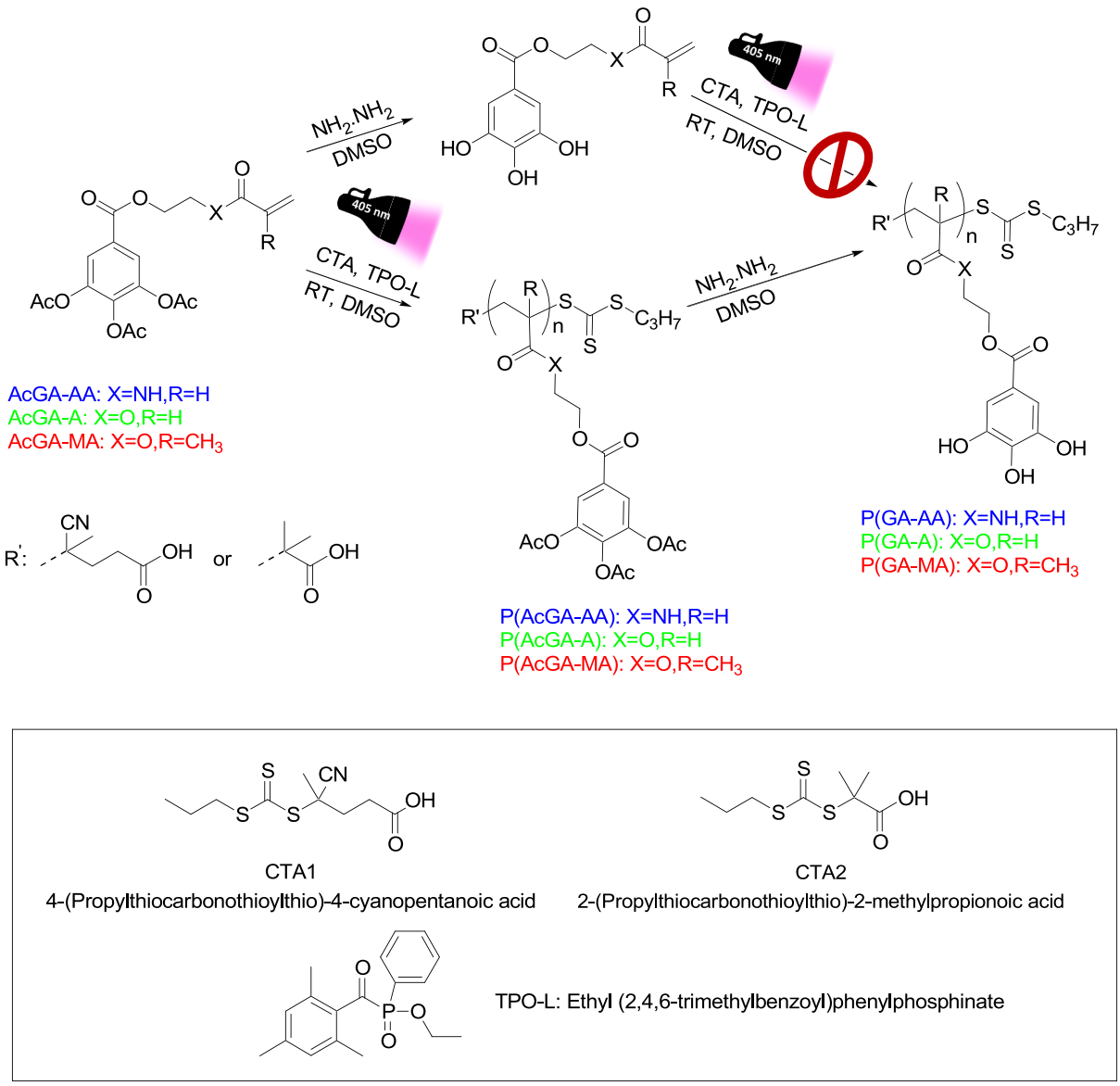
The photo-RAFT polymerization kinetics of AcGA-A using CTA1 and CTA2 are represented by the plot of first-order kinetic *vs* time. As shown in Fig. 2, very high conversion values were recorded, whichever the RAFT agent used (93 % with CTA1 after 85 min, 97 % with CTA2 after 20 min). In comparison, the polymerization using CTA2 was faster than with CTA1 while maintaining a linear first-order kinetic plot. It is worth mentioning the presence of a short induction period when CTA1 was used, that is consistent with the literature reporting the lower reactivity of acrylate monomers when using CTA1 compared to CTA2. Indeed, CTA2 is known to be more compatible with acrylates than methacrylates [34].

In a similar manner, the photo-RAFT polymerization of AcGA-MA using CTA1 was monitored. A good conversion (73 %) was achieved after 7.5 h, with a linear progression of first-order kinetics. On the contrary, progressive deactivation of the polymerization carried out with CTA2 occurs while irradiating the system. Such results agree with the literature confirming the better compatibility of CTA1 to the polymerization of methacrylates in comparison to CTA2. Our previous results also reported a much better photo-RAFT polymerization control of perbenzylated GA-based methacrylate with CTA1 [10].

When CTA1 was used in case of AcGA-AA, uncontrolled polymerization was observed with a relatively high conversions (80 %) after 2 h of irradiation. Then, the conversion no longer progresses. On the contrary, the polymerization of AcGA-AA in the presence of CTA2 was successfully achieved with high conversion (97 %) in a relatively short reaction time (20 min) and a linear first-order kinetic plot was drawn. As a result, CTA2 appears to be a more efficient than CTA1 for the polymerization of AcGA-AA in a controlled manner. This confirms the compatibility of this CTA2 trithiocarbonate RAFT agent for the polymerization of acrylamides on the contrary to CTA1 [34]. It is worth mentioning the presence of a short induction period during the polymerization of AcGA-AA, whichever the RAFT agent used.

^{1}H NMR spectra of pure P(AcGA-AA), P(AcGA-A) and P(AcGA-MA) are shown in Fig. 3. The total disappearance of all vinylic proton signals from residual monomer was observed between 5.4 and 6.4 ppm. In addition, for each case, we observed a low shift of peaks characteristic of aromatic protons (e'), acetyl protons (g' and f') and protons (c' and d') to lower ppm values. Moreover, upon polymerization, new aliphatic main-chain protons appeared below 1 ppm for all the polymers. In case of P(AcGA-MA), a shift of the aliphatic methyl protons peak (b') was observed from 1.8 (Fig. 1) to around 0.75 ppm (Fig. 3). The J-mod ^{13}C NMR spectra of protected GA-based polymers (Fig. S5) also demonstrate the disappearance of peaks characteristic of monomer C=C bond (125 to 137 ppm in Fig. S1) and the appearance of aliphatic main-chain carbon peaks between 15 and 25 ppm. The FTIR spectra of such protected GA-based polymers show another evidence of the disappearance of C = C band around 1620–1640 cm^{-1} (Fig. S2).

After selection of experimental conditions leading to control the photo-RAFT polymerization of such monomers, the SEC curves of polymers shift to lower elution volume with the increase of the conversion and remain monomodal (Figs. 4 and S6). The dispersities and molecular weights estimated by SEC *vs* monomer conversion are also drawn in Fig. 4. Consistent experimental molecular weights estimated



Scheme 2. Synthesis of free GA-based polymers (P(GA-AA), P(GA-A) and P(GA-MA)). Insert: chemical structures of TPO-L and CTAs.

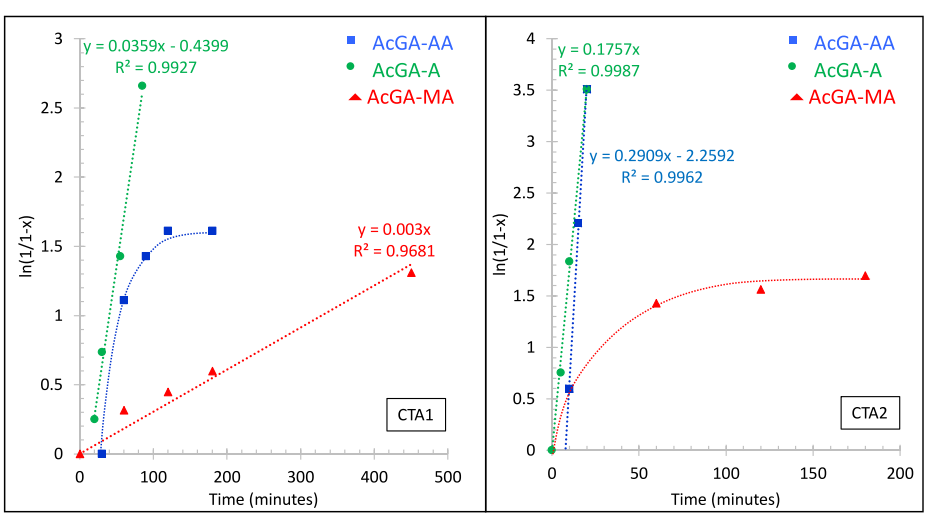


Fig. 2. Semilogarithmic plots *vs* polymerization time during photo-RAFT using CTA1 and CTA2 as chain transfer agents. $[CTA]_0/[TPO\text{-}L]_0 = 10$ and $[M]_0/[CTA]_0 = 200$.

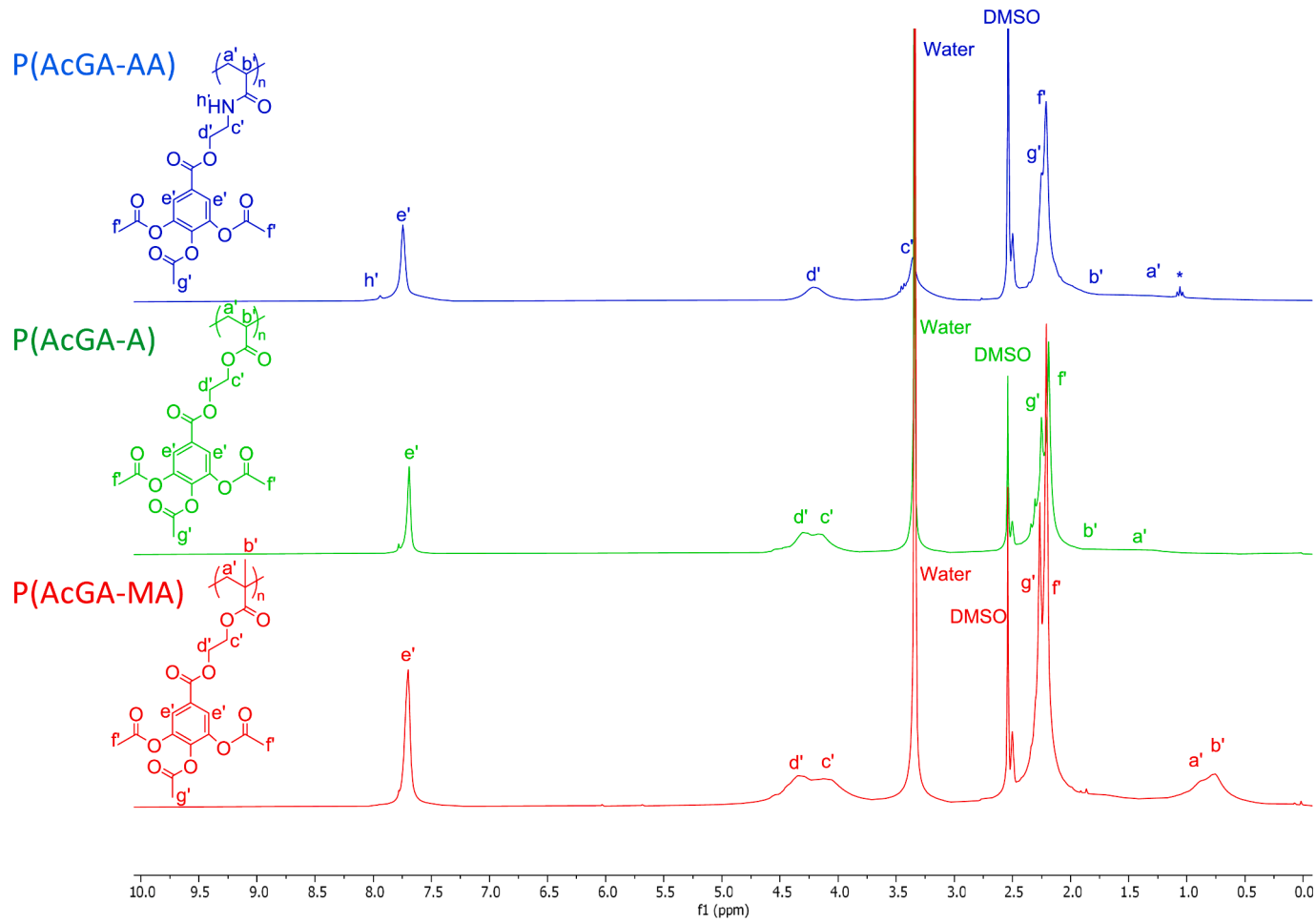


Fig. 3. ^{1}H NMR spectra (DMSO-d_6) of P(AcGA-AA), P(AcGA-A) and P(AcGA-MA).

by both ^{1}H NMR and SEC ($\overline{M}_{nNMR}$ and $\overline{M}_{nSEC}$) agreed with theoretical values ($\overline{M}_{ntheo}$) (Table 1). The low dispersities and linear increases of molecular masses with conversion demonstrate the control of the photo-RAFT of such triacetylated GA-based acrylamide, acrylate and methacrylate using TPO-L as photo-initiator and adequate chain transfer agent. It's worth noting here that experimental conditions enabled the photo-RAFT polymerization of triacetylated AG-based acrylate (AcGA-A) to be controlled whereas no control had previously been obtained when the OH functions of a similar AG-based acrylate were protected by benzyl groups [10].

Finally, thermal characterization of protected polymers was done by DSC (Table 1). The Tg values of P(AcGA-A) was slightly higher than the P(AcGA-MA) (62.8 °C *vs* 60.8 °C). These results are consistent with the previously reported values of perbenzylated GA-based polymers [10]. A significant increase in the value of Tg of P(AcGA-AA) was observed (85.1 °C) due to the hydrogen bonds of the amide function. The smaller size of the protecting acetyl group causes the increase of Tg values compared to the perbenzylated version of the polymers [10].

3.3. Deprotection of acetylated polymers

To obtain free GA-based polymers, a deprotection step of the OH functions of P(AcGA-AA), P(AcGA-A) and P(AcGA-MA) was performed (Scheme 2). Unlike our previous work, the deprotection was herein done carrying out a metal-free deprotection procedure. More precisely, the deprotection of the triacetylated polymers was done using hydrazine hydrate at room temperature for a short reaction time (45 min). This fast method allowed the quantitative hydrolysis of acetate protecting groups for every repeating unit (Scheme 2). Deprotected GA-based polymers thus obtained, P(GA-AA), P(GA-A) and P(GA-MA), exhibit a free GA group on each repeating unit. The success of such deprotection was demonstrated through ^{1}H, J-mod ^{13}C NMR spectra and FTIR analysis. Fig. 5 shows the disappearance of acetyl protons peaks originally located around 2.3 ppm (Fig. 3). Fig. 5 shows also a decrease in the chemical shift of aromatic protons to below 7 ppm (similar to the aromatic protons chemical shift of gallic acid). The appearance of broad peaks above 7.5 ppm (f``), representing the labile phenolic protons was observed.

J-mod ^{13}C NMR spectra (Fig. S7) presented the disappearance of protecting groups (disappearance of peaks of carbons 11–14 in the previous Fig. S5) and the shift of peaks characteristic of aromatic carbons 7, 8 and 9 (to values similar to those of GA). As presented in Fig. S2, FTIR of free-GA based polymers P(GA-AA), P(GA-A) and P(GA-MA) clearly demonstrated the success of such deprotection, through the appearance of broad bands corresponding to OH stretch absorption at 3250 cm^{-1}.

SEC analysis of free GA-polymers were performed in DMSO/$NaNO_3$ eluent (Fig. 6) and results are presented in Table 2. As previously reported [10], the generation of numerous hydrogen bonds between all the phenol parts along the polymeric chain causes the increase of the dispersity values in addition to possible adhesion to the column.

Upon deprotection, the thermal characteristics of free GA-based polymers (Table 2) were studied using DSC analysis. As previously mentioned, numerous hydrogen bonds appear and induce a significant increase in the values of Tg for all the polymers (see Table 1 for Tg of protected polymers), which then range from 127 to 136 °C. Moreover, a very slightly higher Tg value for P(GA-AA) was observed in comparison

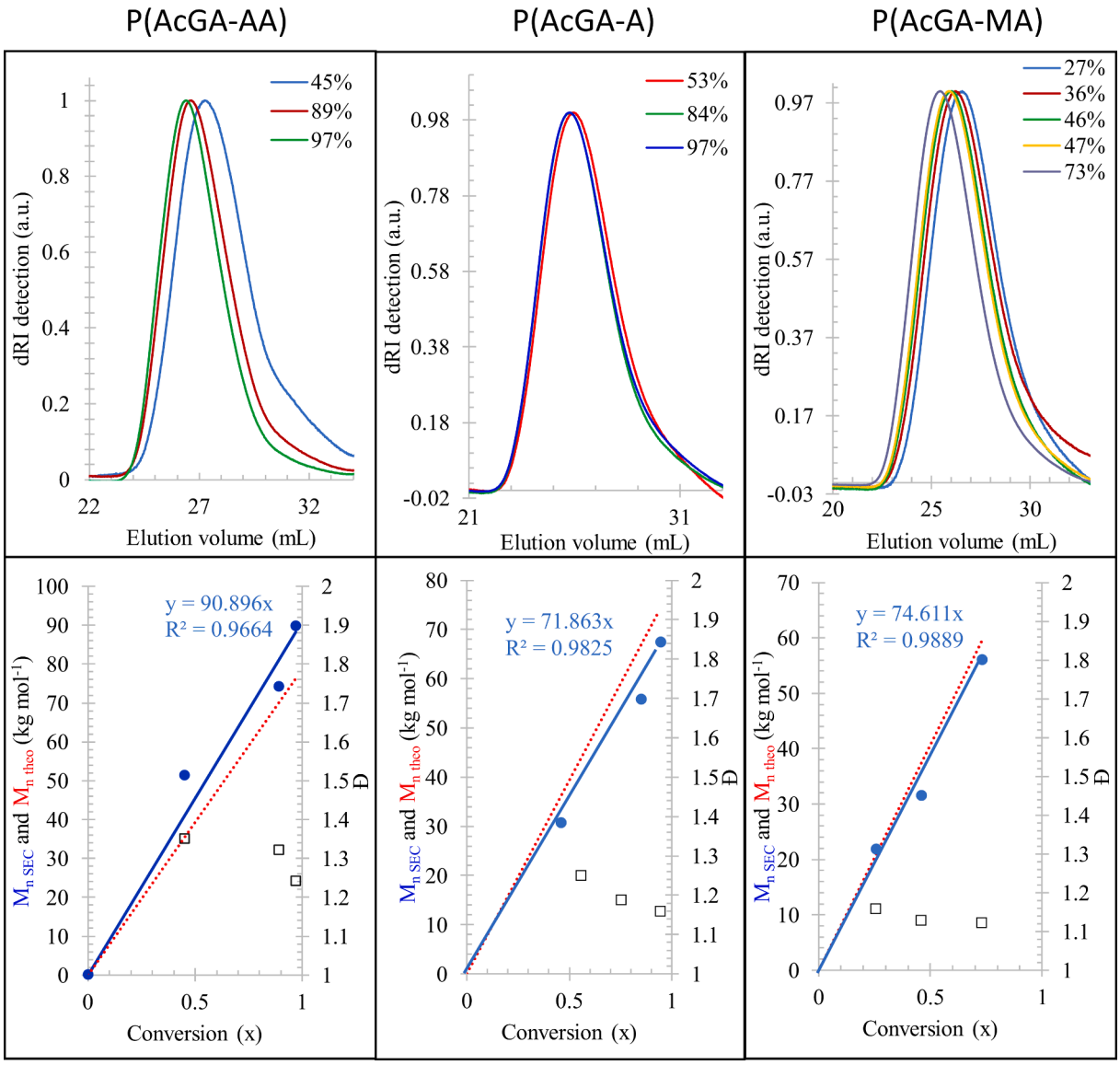


Fig. 4. Top: SEC traces (dRI detection) at various conversion. Down: Plots of $\overline{M}_{nSEC}$ (solid line, •), $\overline{M}_{ntheo}$ (dotted line) and Đ (□) *vs* conversion. Case of the polymerizations of AcGA-AA using CTA2, and AcGA-A / AcGA-MA using CTA1.

Table 1
Photo-RAFT polymerizations of AcGA-AA, AcGA-A and AcGA-MA in DMSO at RT. Initial ratio $[M]_0/[CTA]_0/[TPO\text{-}L] = 200/1/0.1$ was used. Molar masses and glass temperatures.

Monomer	CTA	Time (min)	Conversion (%) [a]	$\overline{M}_{nNMR}$(kg/mol) [b]	$\overline{M}_{nSEC}$(kg/mol) [c]	Đ [c]	T_g (°C)[d]
AcGA-AA	CTA1	120	80	63.2	89.6	1.26	
	CTA2	20	97	76.5	90.5	1.26	85.1
AcGA-A	CTA1	85	93	73.4	67.8	1.35	
	CTA2	20	97	76.5	48.0	1.16	62.8
AcGA-MA	CTA1	450	73	59.8	56.1	1.13	60.8
	CTA2	300	52	42.6	40.2	1.25	

a) Conversion determined by ^{1}H NMR according to equations (1)–(3).
b) $\overline{M}_{nNMR} = \text{conversion} \times M_0 \times 200 + M_{CTA}$, with M_0 the respective molecular weight of monomer unit ($M_0^{AcGA-AA}$: 393.35 g/mol, M_0^{AcGA-A}: 393.33 g/mol, $M_0^{AcGA-MA}$: 408.11 g/mol) and M_{CTA} the respective molecular weight of CTA (M_0^{CTA1}: 277.43 g/mol, M_0^{CTA2}: 238.03 g/mol).
c) $\overline{M}_n$ and dispersity estimated by SEC analysis in DMSO/$NaNO_3$ at 70°C.
d) Glass transition temperature measured using the 2nd heating cycle.

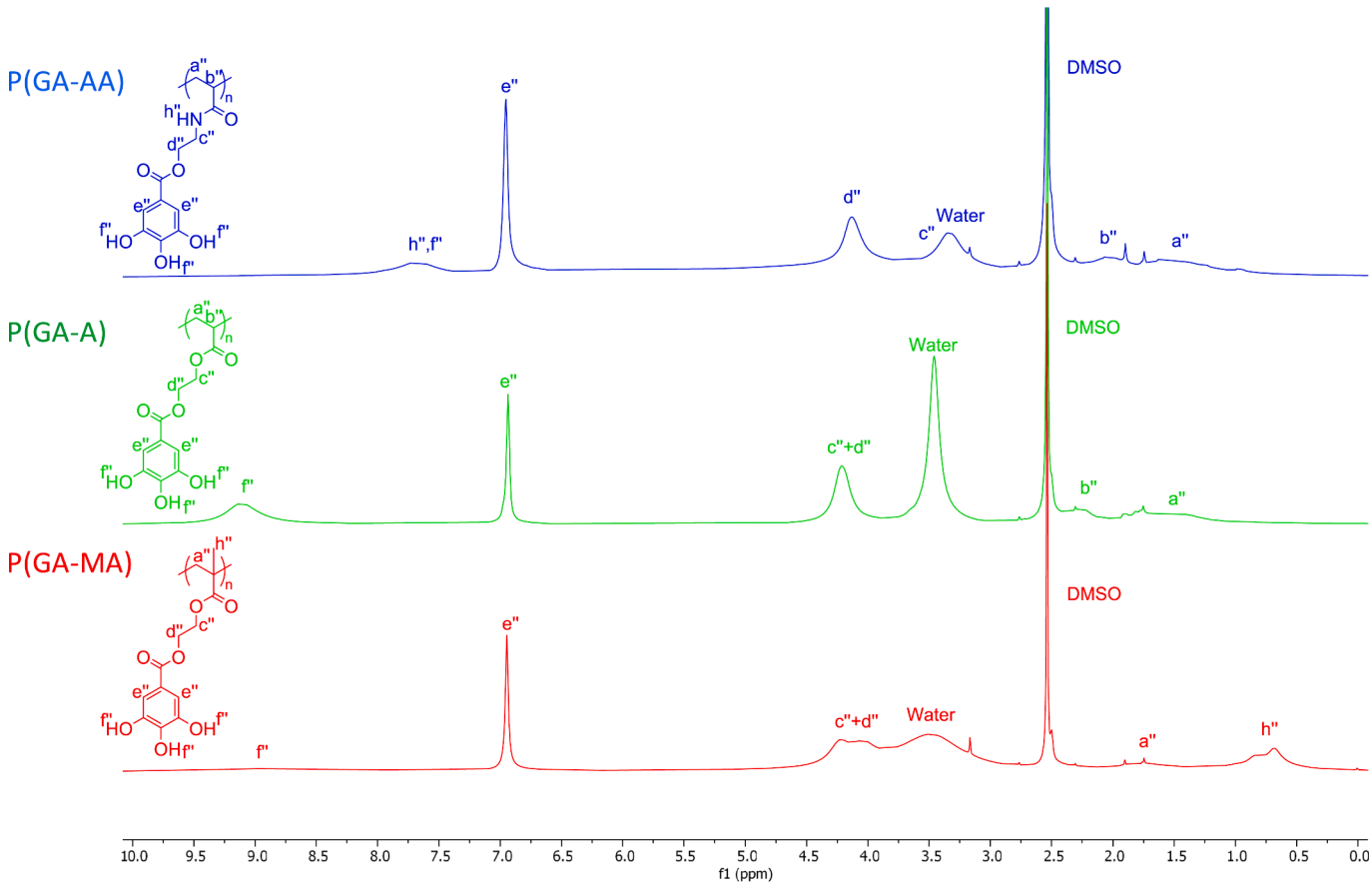


Fig. 5. ^{1}H NMR spectra (DMSO-d_6) of P(GA-AA), P(GA-A) and P(GA-MA).

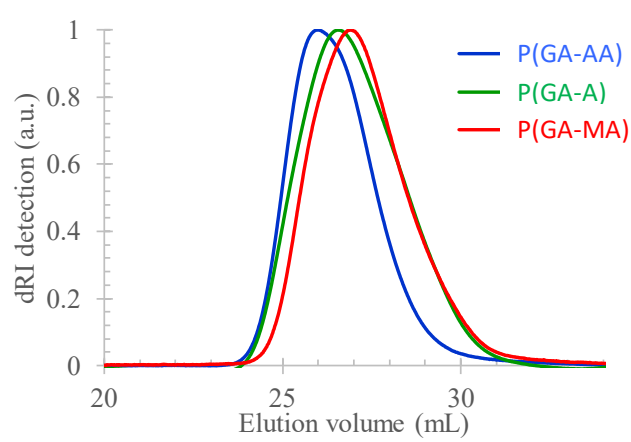


Fig. 6. SEC traces of deprotected polymers. P(AcGA-AA) and P(AcGA-A) were obtained at 97% conversion, when P(AcGA-MA) was produced at 73 % conversion (see Table 1).

Table 2
Molecular weights and glass transition temperatures of free GA-based polymers.

System used to produce triacetylated polymers[a)]	Conv (%)	Free GA-based polymers				
		Name	$\overline{M}_{n\,NMR}$ (kg/mol) [b)]	$\overline{M}_{n\,SEC}$ (kg/mol) [c)]	$Đ$ [c)]	T_g (°C) [d)]
AcGA-AA/CTA2	97	P(GA-AA)	52.0	59.5	1.38	135.3
AcGA-A/CTA2	97	P(GA-A)	50.1	34.0	1.30	129.5 [e)]
AcGA-MA/CTA1	73	P(GA-MA)	41.4	40.1	1.28	127.2 [e)]

a) see Table 1 for experimental conditions.
b) $\overline{M}_{n\,NMR}$ = conversion × M'_0 × 200 + M_{CTA}, with M'_0 the respective molecular weight of deprotected monomer unit (M'^{GA-AA}_0 : 267.24 g/mol, M'^{GA-A}_0 : 268.22 g/mol, M'^{GA-MA}_0 : 282.25 g/mol).
c) $\overline{M}_n$ and dispersity Đ estimated by SEC analysis in DMSO/$NaNO_3$ at 70°C. (SEC traces in Fig. 6).
d) Glass transition temperature measured using the 2nd heating cycle.
e) as reported in Ref. [10].

with that of P(GA-A) due to additional hydrogen bonds coming from the amide function.

3.4. Antioxidant activity of free GA-based polymers

In our previous article [10], the antioxidant activity of free GA-based poly(meth)acrylates (P(GA-A) and P(GA-MA)) against DPPH was demonstrated. Such polymers showed very good antioxidant activity thank to each GA lateral unit, that is one of the best natural antioxidants. Similarly, we performed herein DPPH scavenging assay with the deprotected polyacrylamide P(GA-AA). It was easily observed that the violet colour of DPPH radicals changed into yellow colour of the reduced form of DPPH. As demonstrated in Fig. S8A, the absorbance decreases with the increase in the concentration of P(GA-AA) to reach a plateau when the concentration exceeds 50 μM (not shown). The value of critical concentration (C_{50}, 50 % of reduction of the initial DPPH) registered for P(GA-AA) was similar (17 μM) to those reported for P(GA-A) and P(GA-MA) *vs* 8 μM in case of gallic acid [10] (Table S1). In a similar behaviour,

the percentage of scavenged DPPH radicals increase proportionally in a linear manner. The lower antioxidant properties of free GA-based polymers *vs* GA were attributed to the steric hindrance from the polymeric chains leading to lower accessibility of adjacent GA lateral substituents as already reported [35].

Additionally, the effect of the antioxidant GA moiety carried by P(GA-AA) on the oxidation of methyl linoleate was evaluated. Usually, the presence of radicals from AIBN and oxygen with methyl linoleate can cause its oxidation. Such oxidation can be inhibited by the presence of an antioxidant molecule such as GA. Subsequently, to evaluate the antioxidation power of P(GA-AA), the consumption of oxygen in the reactor was monitored throughout the experiment time. The reduction of the O_2 pressure ($\Delta P = P_0(O_2) - P_{2.5\,h}(O_2)$, where $P_0(O_2)$ and $P_{2.5\,h}(O_2)$ are the initial and after 2.5 h oxygen pressures, respectively) in the reactor *vs* time in the presence of P(GA-AA) (Fig. S8B), was also compared to that of P(GA-A) and P(GA-MA) [10]. As shown, a linear evolution of the pressure variation was also recorded for P(GA-AA). A slightly lower inhibition percentage (50 %) was recorder for P(GA-AA), as compared to the previously determined values (55 %) for the poly(meth)acrylates (Table S1).

3.5. Antibacterial activity of free GA-based polymers

GA, as many other phenolic compounds, has shown antibacterial activity against various bacterial strains and minimal inhibitory concentrations (MIC, known to be the lowest concentration to completely inhibit bacterial growth) were determined. Unfortunately, such MIC values depend on the intrinsic characteristics of the bacteria and the experimental conditions such as pH, temperature, media composition and assay methods for instance. For instance, on one hand, MIC values for GA were reported to be 125 mg/L (i.e. 6.6 10^{-4} mol/L) against *E. coli* [36] and 12.5 mg/L (i.e. 6.6 10^{-5} mol/L) against *S. aureus* [37]. On another hand, MIC for GA was found equal to 500 mg/mL (i.e. 2.65 mol/L) against *P. aeruginosa*, 1500 mg/mL (i.e. 7.97 mol/L) against *E. coli* and 1750 mg/mL (i.e. 9.30 mol/L) against *S. aureus* [38]. The variations in these MIC values highlight the specificity of GA for each bacterial strain.

The mechanism of GA action is still unknown, even though many proposals have been reported. In this regard, it seems that Gram-negative bacteria are more sensitive to the bactericidal mechanism of GA than Gram-positive bacteria. For instance, it was found that GA interact with the bacteria membranes leading to irreversible alteration of the membrane permeability, physicochemistry and hydrophobicity [38,39]. Zeta potential measurements demonstrated that the bacterial cells become less negatively charged after exposition to GA.

The antibacterial activity of free GA-based polymers (P(GA-AA), P(GA-A) and P(GA-MA) was thus herein estimated, and compared with that of GA by determining the MIC values. In the present paper, MIC was expressed in molar concentration of GA (molecules in case of GA, or lateral substituents in case of polymers), whereas the MIC is usually expressed in weight concentration (mg/mL). However, considering the molar masses of GA and of free GA-based polymers, it is not reasonable to compare MIC values in mg/ml. MICs were herein measured against *E. coli* and *P. aeruginosa*, two common Gram-negative bacterial strains and against two common Gram-negative ones (*Streptococcus* sp. and *S. aureus*). Additionally, two clinical bacteria (*K. aerogenes* (Gram-negative) and *S. aureus*) extracted from otitis externa were also used. As confirmed in Fig. 7, the antibacterial activity was influenced by both the bacteria strain and the polymer chemistry, although results suggest that Gram-negative bacteria are more sensitive to the bactericidal mechanism of action of gallic acid than Gram-positive bacteria. The different values for MIC obtained for each bacteria strain are mainly different, based on the fact that the mechanisms of action are diverse. MIC values in case of free GA-based polymers are undoubtedly higher than that of GA, the latter agreeing with values already reported [36]. Higher MIC values in case of polymers mean that the antibacterial activity of such polymers is lower than that of GA, and may be attributed to the steric hindrance from the polymeric chain leading to lower accessibility of adjacent GA units, as in case of antioxidant properties [10,35]. However, GA is completely metabolized under physiological conditions within 6 h [40], which might impede the antibacterial effect. The polymerization of GA-derived monomer might extend GA elimination half-time (the time needed to reduce by half the concentration of a drug in the human body), and potentially its antibacterial effect.

4. Conclusion

Polyacrylamide, polyacrylate and polymethacrylate carrying one GA moiety as lateral substituent in each monomer unit was herein reported using a controlled scheme, from the synthesis of protected monomer, photo-RAFT polymerization to deprotection steps. Such free GA-based polymers ((P(GA-AA), P(GA-A) and P(GA-MA)) exhibit antioxidant and antibacterial activities. Certainly, these properties are lower than that of GA, that may be attributed to the steric hindrance from the polymeric chain leading to lower accessibility of adjacent GA units. Such properties could be enhanced by performing copolymerization of each triacetylated GA-based monomer with the commercial parent acrylamide/(meth)acrylate (HEAA, HEA and HEMA) for instance. Finally, these polymers could be interesting for intrinsic antimicrobial packaging, in which antimicrobial agents are incorporated directly on the polymer chain to suppress the activities of specific microorganisms.

CRediT authorship contribution statement

Ali Khalil: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Carmen Moya Lopez:** Writing – original draft, Formal analysis, Data curation. **Francisco Carlos Pérez-Martínez:** Data curation. **Christine Gerardin-Charbonnier:** Supervision, Project administration, Funding acquisition, Conceptualization. **Khalid Ferji:** Writing – review & editing, Supervision, Project administration. **Carlos Alonso Moreno:** Writing – original draft, Validation. **Jean-Luc**

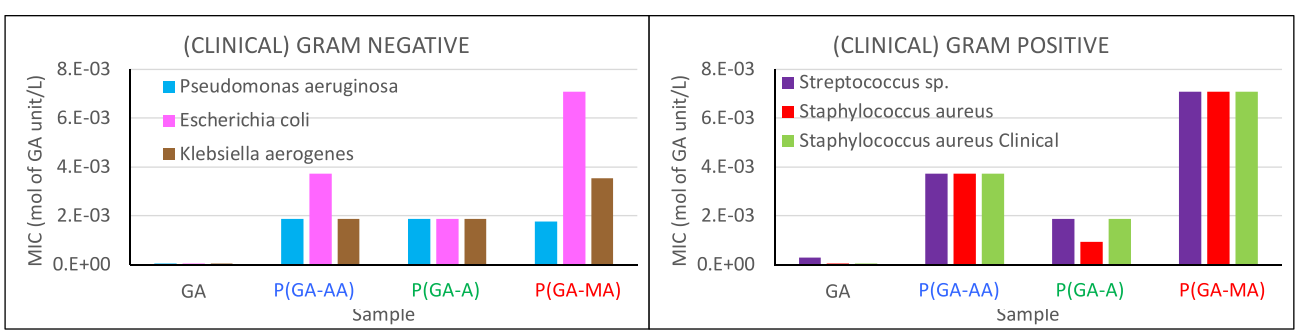


Fig. 7. MIC values (mol/L) gallic acid (GA) and free GA-based polymers (P(GA-AA), P(GA-A) and P(GA-MA) in the presence of several Gram-negative and Gram-positive bacterial strains.

Six: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors thank (i) the "Impact Biomolecules" project of the "Lorraine Université d'Excellence" (Investissements d'avenir, ANR-15-IDEX-04-LUE) funds and (ii) the Energy Mechanics Processes Products Research Department of the Université de Lorraine for financial supports. The authors also thank Olivier Fabre (LCPM) for NMR characterization, Alexandre Collar (LCPM) for SEC characterization, Jean-Claude Sivault (LCPM) for homemade UV−vis LED lamp development, and Béatrice Georges (LERMAB) for antioxidant tests.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113022.

References

[1] S.G.S. Choubey, L.R. Varughese, V. Kumar, A.K. Sharma, V. Beniwal, Probing gallic acid for its broad Spectrum applications, Mini Rev. Med. Chem. 18 (15) (2018) 1283–1293, https://doi.org/10.2174/1389557518666180330114010.
[2] L. Panzella, Natural phenolic compounds for health, food and cosmetic applications, Antioxidants (Basel) 9 (5) (2020), https://doi.org/10.3390/antiox9050427.
[3] R. Mark, X. Lyu, J.J.L. Lee, R. Parra-Saldívar, W.N. Chen, Sustainable production of natural phenolics for functional food applications, J. Funct. Foods 57 (2019) 233–254, https://doi.org/10.1016/j.jff.2019.04.008.
[4] S. Keyvani-Ghamsari, M. Rahimi, K. Khorsandi, An update on the potential mechanism of gallic acid as an antibacterial and anticancer agent, Food Sci. Nutr. 11 (10) (2023) 5856–5872, https://doi.org/10.1002/fsn3.3615.
[5] M. Saeedi, M.R. Moghbeli, O. Vahidi, Chitosan/glycyrrhizic acid hydrogel: Preparation, characterization, and its potential for controlled release of gallic acid, Int. J. Biol. Macromol. 231 (2023) 123197, https://doi.org/10.1016/j.ijbiomac.2023.123197.
[6] T. Thanyacharoen, P. Chuysinuan, S. Techasakul, P. Nooeaid, S. Ummartyotin, Development of a gallic acid-loaded chitosan and polyvinyl alcohol hydrogel composite: release characteristics and antioxidant activity, Int. J. Biol. Macromol. 107 (2018) 363–370, https://doi.org/10.1016/j.ijbiomac.2017.09.002.
[7] M.F. Aldawsari, F.K. Alkholifi, A.I. Foudah, M.H. Alqarni, A. Alam, M.A. Salkini, S. H. Sweilam, Gallic-acid-loaded PLGA Nanoparticles: a promising transdermal drug delivery system with antioxidant and antimicrobial agents, Pharmaceuticals 16 (2023).
[8] A. Sheikh, S.A. Hazari, N. Molugulu, S.A. Alshehri, S. Wahab, A. Sahebkar, P. Kesharwani, Hyaluronic acid engineered gallic acid embedded chitosan nanoparticle as an effective delivery system for treatment of psoriasis, Environ. Res. 238 (2023) 117086, https://doi.org/10.1016/j.envres.2023.117086.
[9] N. Chunshom, P. Chuysinuan, T. Thanyacharoen, S. Techasakul, S. Ummartyotin, Development of gallic acid/cyclodextrin inclusion complex in freeze-dried bacterial cellulose and poly (vinyl alcohol) hydrogel: controlled-release characteristic and antioxidant properties, Mater. Chem. Phys. 232 (2019) 294–300, https://doi.org/10.1016/j.matchemphys.2019.04.070.
[10] A. Khalil, C. Gerardin-Charbonnier, H. Chapuis, K. Ferji, J.-L. Six, Original bio-based antioxidant poly(meth)acrylate from gallic acid-based Monomers, ACS Sustain. Chem. Eng. 9 (34) (2021) 11458–11468, https://doi.org/10.1021/acssuschemeng.1c03607.
[11] S.J. Uddin, M. Afroz, S.M.N.K. Zihad, M.S. Rahman, S. Akter, I.N. Khan, S.M.S. Al-Rabbi, R. Rouf, M.T. Islam, J.A. Shilpi, et al., A systematic review on anti-diabetic and Cardioprotective potential of gallic acid: a widespread Dietary phytoconstituent, Food Rev. Intl. 38 (4) (2022) 420–439, https://doi.org/10.1080/87559129.2020.1734609.
[12] S. Shabani, Z. Rabiei, H. Amini-Khoei, Exploring the multifaceted neuroprotective actions of gallic acid: a review, Int. J. Food Prop. 23 (1) (2020) 736–752, https://doi.org/10.1080/10942912.2020.1753769.
[13] N. Kahkeshani, F. Farzaei, M. Fotouhi, S.S. Alavi, R. Bahramsoltani, R. Naseri, S. Momtaz, Z. Abbasabadi, R. Rahimi, M.H. Farzaei, et al., Pharmacological effects of gallic acid in health and diseases: a mechanistic review, Iran J. Basic Med. Sci. 22 (3) (2019) 225–237, https://doi.org/10.22038/ijbms.2019.32806.7897.
[14] W. Pasanphan, S. Chirachanchai, Conjugation of gallic acid onto chitosan: an approach for green and water-based antioxidant, Carbohydr. Polym. 72 (1) (2008) 169–177, https://doi.org/10.1016/j.carbpol.2007.08.002.
[15] M. Xie, B. Hu, Y. Wang, X. Zeng, Grafting of gallic acid onto chitosan enhances antioxidant activities and alters rheological properties of the Copolymer, J. Agric. Food Chem. 62 (37) (2014) 9128–9136, https://doi.org/10.1021/jf503207s.
[16] J. Liu, H. Yong, Y. Liu, R. Bai, Recent advances in the preparation, structural characteristics, biological properties and applications of gallic acid grafted polysaccharides, Int. J. Biol. Macromol. 156 (2020) 1539–1555, https://doi.org/10.1016/j.ijbiomac.2019.11.202.
[17] U.G. Spizzirri, F. Iemma, F. Puoci, G. Cirillo, M. Curcio, O.I. Parisi, N. Picci, Synthesis of antioxidant Polymers by grafting of gallic acid and catechin on gelatin, Biomacromolecules 10 (7) (2009) 1923–1930, https://doi.org/10.1021/bm900325t.
[18] B. Huang, Z. Zhang, N. Ding, B. Wang, G. Zhang, P. Fei, Investigation of the pectin grafting with gallic acid and propyl gallate and their antioxidant activities, antibacterial activities and fresh keeping performance, Int. J. Biol. Macromol. 190 (2021) 343–350, https://doi.org/10.1016/j.ijbiomac.2021.08.219.
[19] S. Yan, Q. Wang, Y. Li, B. Qi, Gallic acid-functionalized soy protein-based multiple cross-linked hydrogel: mechanism analysis, physicochemical properties, and digestive characteristics, Food Chem. 433 (2024) 137290, https://doi.org/10.1016/j.foodchem.2023.137290.
[20] A. Guchait, S. Sharma, S. Chattopadhyay, T. Mondal, Synthesis of gallic acid-grafted epoxidized natural rubber and its role in self-healable flexible temperature sensors, Soft Matter 20 (1) (2024) 178–191, https://doi.org/10.1039/D3SM01367G.
[21] V. Marturano, A. Marotta, S.A. Salazar, V. Ambrogi, P. Cerruti, Recent advances in bio-based functional additives for polymers, Prog. Mater Sci. 139 (2023) 101186, https://doi.org/10.1016/j.pmatsci.2023.101186.
[22] A. Rigoussen, P. Verge, J.-M. Raquez, P. Dubois, Natural phenolic antioxidants as a source of biocompatibilizers for immiscible Polymer blends, ACS Sustain. Chem. Eng. 6 (10) (2018) 13349–13357, https://doi.org/10.1021/acssuschemeng.8b02999.
[23] D. Ikkene, A.A. Arteni, M. Ouldali, G. Francius, A. Brûlet, J.-L. Six, K. Ferji, Direct access to Polysaccharide-based vesicles with a tunable membrane thickness in a Large concentration window via Polymerization-induced self-assembly, Biomacromolecules 22 (7) (2021) 3128–3137, https://doi.org/10.1021/acs.biomac.1c00569.
[24] D. Ikkene, J.-L. Six, K. Ferji, Progress in aqueous dispersion RAFT PISA, Eur. Polym. J. 188 (2023) 111848, https://doi.org/10.1016/j.eurpolymj.2023.111848.
[25] V.L. Romero Castro, B. Nomeir, A.A. Arteni, M. Ouldali, J.-L. Six, K. Ferji, Dextran-coated latex Nanoparticles via photo-RAFT mediated Polymerization induced self-assembly, Polymers 13 (2021).
[26] D. Ikkene, A.A. Arteni, C. Boulogne, J.-L. Six, K. Ferji, Multicompartment vesicles: a key intermediate structure in Polymerization-induced self-assembly of graft Copolymers, Macromolecules 55 (11) (2022) 4268–4275, https://doi.org/10.1021/acs.macromol.2c00561.
[27] D. Ikkene, A.A. Arteni, M. Ouldali, J.-L. Six, K. Ferji, Self-assembly of amphiphilic copolymers containing polysaccharide: PISA versus nanoprecipitation, and the temperature effect, Polym. Chem. 11 (29) (2020) 4729–4740, https://doi.org/10.1039/D0PY00407C.
[28] D. Ikkene, A.A. Arteni, H. Song, H. Laroui, J.L. Six, K. Ferji, Synthesis of dextran-based chain transfer agent for RAFT-mediated polymerization and glyco-nanoobjects formulation, Carbohydr. Polym. 234 (2020) 115943, https://doi.org/10.1016/j.carbpol.2020.115943.
[29] K. Ferji, P. Venturini, F. Cleymand, C. Chassenieux, J.-L. Six, In situ glyco-nanostructure formulation via photo-polymerization induced self-assembly, Polym. Chem. 9 (21) (2018) 2868–2872, https://doi.org/10.1039/C8PY00346G.
[30] P. Lertturongchai, M.I.A. Ibrahim, A. Durand, P. Sunintaboon, K. Ferji, Synthesis of thermoresponsive Copolymers with tunable UCST-type phase transition using aqueous photo-RAFT Polymerization, Macromol. Rapid Commun. 41 (9) (2020) 2000058, https://doi.org/10.1002/marc.202000058.
[31] F. Liu, Y. Long, Q. Zhao, X. Liu, G. Qiu, L. Zhang, Q. Ling, H. Gu, Gallol-containing homopolymers and block copolymers: ROMP synthesis and gelation properties by metal-coordination and oxidation, Polymer 143 (2018) 212–227, https://doi.org/10.1016/j.polymer.2018.04.016.
[32] J.M. Nasrullah, S. Raja, K. Vijayakumaran, R. Dhamodharan, A practical route for the preparation of poly(4-hydroxystyrene), a useful photoresist material, J. Polym. Sci. A Polym. Chem. 38 (3) (2000) 453–461, https://doi.org/10.1002/(SICI)1099-0518(20000201)38:3.
[33] P. Seguí, J.J. Aguilera-Correa, E. Domínguez-Jurado, C.M. Sánchez-López, R. Pérez-Tanoira, A.V. Ocaña, J.A. Castro-Osma, J. Esteban, A. Marcilla, C. Alonso-Moreno, et al., A novel bis(pyrazolyl)methane compound as a potential agent against gram-positive bacteria, Sci. Rep. 11 (1) (2021) 16306, https://doi.org/10.1038/s41598-021-95609-z.
[34] Y. Lee, C. Boyer, M.S. Kwon, Photocontrolled RAFT polymerization: past, present, and future, Chem. Soc. Rev. 52 (9) (2023) 3035–3097, https://doi.org/10.1039/D1CS00069A.

[35] K. Zhan, H. Ejima, N. Yoshie, Antioxidant and adsorption properties of bioinspired phenolic polymers: a Comparative study of catechol and gallol, ACS Sustain. Chem. Eng. 4 (7) (2016) 3857–3863, https://doi.org/10.1021/acssuschemeng.6b00626.
[36] A.-Z. Shm, Antibacterial activities of gallic acid and gallic acid methyl ester on methicillin-resistant Staphylococcus aureus, J. Am. Sci. 8 (2) (2012) 7–12.
[37] J.R.A.R.C. Silva, A.C. Pinto, A.C.F. Amaral, Cytotoxicity and antibacterial studies of iridoids and phenolic compounds isolated from the latex of Himatanthus sucuuba, Afr J Biotechnol 9 (43) (2010) 7357–7360, https://doi.org/10.5897/AJB10.345.
[38] A.F. Borges, Saavedra Carla, J. Maria, Simões, Manuel antibacterial activity and mode of action of ferulic and gallic acids against pathogenic bacteria, Microb. Drug Resist. 19 (4) (2013) 256–265, https://doi.org/10.1089/mdr.2012.0244.
[39] M. Takó, E.B. Kerekes, C. Zambrano, A. Kotogán, T. Papp, J. Krisch, C. Vágvölgyi, Plant phenolics and phenolic-enriched Extracts as antimicrobial agents against food-contaminating microorganisms, In *Antioxidants* 9 (2020).
[40] L. Zong, M. Inoue, M. Nose, K. Kojima, N. Sakaguchi, K. Isuzugawa, T. Takeda, Y. Ogihara, Metabolic fate of gallic acid orally administered to rats, Biol Pharm Bull 22 (3) (1999) 326–329, https://doi.org/10.1248/bpb.22.326.